



◁ COLLIDING CONTINENTS

The High Atlas mountains to the north of Ouarzazate were formed less than 50 million years ago when the African and Eurasian Plates started colliding.

▷ FIERY CRACKS

The Erta Ale volcano lies between the splitting sides of the African and Arabian plates.



▷ ACTIVE VOLCANO

Oldoinyo Lengai in Tanzania is one of many erupting volcanoes along the widening crack in the African Plate that forms the Great Rift Valley. It is unique in that it erupts natrocarbonatite lava.



▷ PARCHED LAND

Almost all of northern Africa is covered by the huge Sahara Desert. Half of the Sahara receives less than 1 in (2.5 cm) of rain per year—the rest gets less than 4 in (10 cm).

In time, volcanic activity forced cracks to open in Pangaea, and as a result Africa split from Europe and South America and started its own journey. Madagascar had split from Gondwana about 20 million years earlier.

A splitting continent

Around 25–22 million years ago, a big crack began to split east Africa in two as the tectonic plates began to pull either side apart and plumes of hot magma thrust their way up from underneath. The widening crack became one of the world's most dramatic geographical features—the Great Rift Valley—which runs over 3,700 miles (6,000 km) from the Dead Sea in Jordan to Mozambique. North of Ethiopia, the rift is splitting the Arabian-Nubian Craton apart and has already gaped wide enough to allow the Indian Ocean

to flood in and create both the Red Sea and the Gulf of Aden. In about 10 million years' time, the two sides of the Great Rift Valley will have pulled far enough apart to eventually open up a new ocean right through the middle of East Africa.

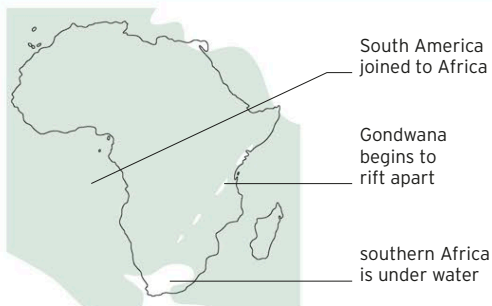
Africa dries out

Africa's location straddling the tropics has created three huge deserts—the Sahara in the north, along the Tropic of Cancer, and the Namib and Kalahari in the south along the Tropic of Capricorn. The Namib is the world's oldest desert and formed 80 million years ago. The Sahara is the world's largest hot desert, yet developed less than 2.5 million years ago—much of it was green until a dramatic climatic shift just 5,000 years ago scorched the north of Africa dry.

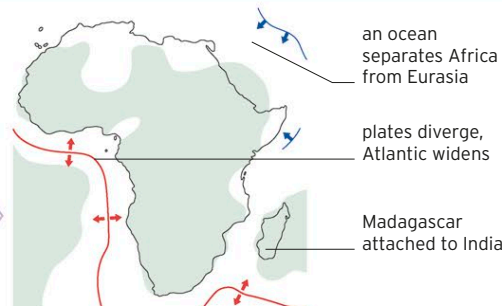
Africa was still attached to South America when Madagascar split away

BUILDING THE AFRICAN CONTINENT

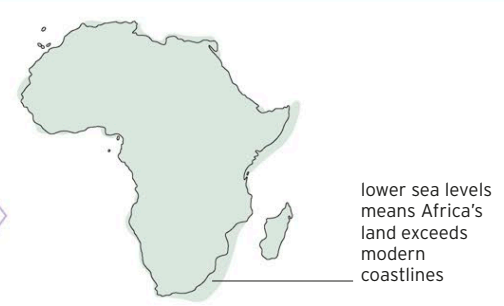
KEY — Convergent boundary — Divergent boundary



237 MILLION YEARS AGO The ancient cratons that form the core of Africa today are locked deep within the supercontinent of Pangaea. Africa is sandwiched between South America and Asia.



94 MILLION YEARS AGO Africa splits away from South America, and as South America drifts west, the South Atlantic opens up between them. Meanwhile, an ocean separates Africa from Eurasia.



18,000 YEARS AGO Africa moves north to collide with Eurasia again, squeezing the ocean separating the continents out of existence, but it retains its separate identity as a continent.